Highlights

How Should a Robot Ask? Comparing Active Questioning Strategies for Learning User Preferences in Household Rearrangement

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- Three complementary questioning strategies reveal distinct accuracy–agency trade-offs
- Simulation across 102 episodes shows category-level questioning doubles unseen-object accuracy
- User study shows individual differences in organizing expertise moderate strategy effectiveness

How Should a Robot Ask? Comparing Active Questioning Strategies for Learning User Preferences in Household Rearrangement

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ABSTRACT

Household tidying is a deeply personal activity shaped by individual habits, mental categories, and tacit organizing principles that users rarely articulate unprompted. When an intelligent agent assists with rearrangement, the central challenge is not *what* to learn but *how to ask*—whether to query individual object placements, elicit category-level rules, or confirm patterns inferred from prior evidence. Each questioning approach embodies a different assumption about users' ability and willingness to articulate preferences at varying levels of abstraction. We identify three complementary strategies operating at different granularities: **Action-Oriented** (AO) questioning targets individual placements, **Preference Eliciting** (PE) uncovers category-level organizing principles, and **Preference Induction** (PI) confirms rules inferred from accumulated evidence. We implement these strategies within an LLM-based conversational agent and investigate them through two studies. Study 1, a large-scale simulation across 102 household episodes at question budgets 0–6, establishes that structured questioning dramatically outperforms both no-question baselines and unstructured LLM querying, with PE achieving up to 71.9% unseen-object accuracy versus 42.6% without questioning. Study 2, a controlled user study currently underway, is designed to examine how strategies affect perceived control, cognitive load, and satisfaction. Guided by the simulation findings, we predict that the accuracy advantage of PE comes at the cost of reduced user sense of agency, and that individual differences in organizing expertise moderate which strategy users find most effective. We discuss theoretical implications for preference communication in human-agent interaction and offer design recommendations for conversational agents that actively learn from users.

1. Introduction

Organizing a home is a skill shaped by personal habit. Two people sharing the same kitchen may sort identical objects along entirely different principles—one by function ("cooking tools near the stove"), another by frequency of use ("daily items at eye level"), a third by aesthetics ("everything visible must match in colour"). These principles are rarely written down or easily verbalized; they emerge from years of practice and tend to feel obvious to the person who holds them. Yet for an intelligent agent tasked with assisting in household rearrangement, the task is precisely to surface these tacit preferences—quickly, unobtrusively, and correctly—before placing objects on the user's behalf.

The central interaction design challenge is not whether to ask, but *how*. An agent operating under a limited question budget must choose between asking about specific objects ("Where should the wireless speaker go?"), eliciting category-level rules ("How do you usually organize electronic accessories?"), or confirming patterns it has inferred from earlier answers ("I've noticed you place books on the display shelf—is that a general rule?"). These choices are not merely technical: they reflect assumptions about how users represent their own preferences, what abstraction levels they can comfortably reason about, and how much control they expect to retain over the organizing decisions.

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Prior work on conversational agents has shown that the form of a question—not just its content—significantly shapes the quality of the answer it elicits and the user's experience of the interaction (**?**). In task-oriented dialogue, the tension between task efficiency and relational quality mirrors the tension between information-dense category questions and simpler object-level queries (**?**). Yet despite this established sensitivity to dialogue strategy, the question of how intelligent agents should ask for *personal preferences*—especially at different granularities—has not been systematically studied.

This paper addresses that gap by investigating three active questioning strategies—*Action-Oriented* (AO), *Preference Eliciting* (PE), and *Preference Induction* (PI)—that differ in the cognitive demands they place on users and the type of preference knowledge they target. We pursue four research questions:

RQ1: How does questioning strategy affect task performance (placement accuracy for seen and unseen objects) across different question budgets?

RQ2: Does the structure of the questioning strategy—as opposed to simply asking questions—provide measurable benefits over unstructured LLM-driven dialogue?

RQ3: How do the strategies differ in their effects on perceived control, cognitive load, satisfaction, and trust in the agent?

RQ4: Do individual characteristics—specifically, users' prior organizing expertise—moderate the relative effectiveness of different strategies?

To address RQ1 and RQ2 at scale, we conduct a large simulated evaluation (Study 1) using LLM-grounded oracles across 102 annotated household episodes. To address RQ3 and RQ4, we conduct a controlled user study (Study 2) with real participants interacting through a conversational interface. The convergence and divergence between simulation and user study results is itself informative: it reveals which aspects of preference communication can be approximated computationally and which require human participants to observe.

Our contributions are:

- A **taxonomy of three active questioning strategies** for preference learning in household rearrangement, formalized along the dimensions of target granularity, cognitive demand, and generalization scope.
- An **LLM-based agent implementation** with strategy-specific proposers, structured state tracking, and a multi-step placement planner, enabling rigorous per-strategy evaluation.
- **Simulation findings** across 102 episodes demonstrating that category-level (PE) questioning nearly doubles unseen-object accuracy over no-question baselines, and that structured strategies consistently outperform unstructured LLM questioning.
- A **controlled user study** designed to characterize the accuracy–agency trade-off across strategies and test whether organizing expertise moderates their relative effectiveness, with implications for adaptive strategy selection.

2. Related Work

2.1. Dialogue Strategy in Human-Agent Interaction

How an agent structures its questions has well-documented consequences for user behavior and experience. **?** compared topic-led and task-led conversation strategies in service chatbots (n=35), finding that topic-led approaches enhance perceived anthropomorphism and hedonic quality while task-led designs improve pragmatic outcomes—a tension that maps directly onto our comparison of category-level (PE) and object-level (AO) questioning. **?** showed that conversation-based questioning outperforms form-based questioning for eliciting detailed user responses, with credibility as a significant mediator; their 2×3 design underscores that the *form* of the question—not just its content—shapes what users reveal.

These findings motivate our core research design: rather than comparing fixed-content questions, we compare qualitatively different questioning strategies—strategies that differ in what cognitive operations

they demand of users. The existing literature demonstrates that such differences matter; it does not yet tell us which strategy granularity best serves the specific challenge of preference elicitation for personalized physical tasks.

2.2. Users' Ability to Articulate Organizing Preferences

A recurring finding in the literature on preference elicitation is that users vary considerably in their capacity to articulate abstract preferences on demand. **?** found that users of robot home assistants differed substantially in their ability to verbalize organizing rules, with some users preferring to demonstrate preferences through specific examples while others readily articulated general principles. **?** demonstrated that users' experience level with LLM interfaces significantly affects their perception of an agent's communicative competence, implying that strategy effectiveness may interact with user expertise.

These observations motivate RQ4: rather than expecting a single strategy to be universally optimal, we treat organizing expertise as a theoretically grounded moderating variable. No prior work has directly examined how this user characteristic interacts with preference-elicitation strategy in the context of physical task assistance—a gap that our user study (Study 2) is designed to close.

2.3. Active Learning and Preference Elicitation in Interactive Systems

Active preference learning—where a system strategically selects which information to request—has been studied in recommendation systems, Bayesian experimental design, and robot learning from demonstration. **?** developed a conversational recommender that matches interaction modes to users' decision-making styles (n=62), demonstrating that adapting the interaction modality to user cognition improves both satisfaction and decision quality. The underlying principle—that the granularity of preference queries should match the user's current representation of their preferences—is central to our investigation.

The shift from passive to interactive AI (**?**) provides theoretical context: rather than inferring preferences from observed behavior alone, our agent actively shapes the information it receives through structured dialogue. In household robotics, TidyBot (**?**) demonstrated that LLM-based generalization from a small number of demonstrations can produce effective placement plans, establishing that preference generalization is feasible. Our work asks the complementary question: what questioning *strategy* enables the agent to gather the most useful information within a fixed budget?

2.4. Proactivity, Control, and Trust in Agent Systems

Our three strategies differ not only in their information-gathering efficiency but in how they position the user relative to the agent's reasoning. AO questioning positions the user as the authority ("tell me where this goes"); PE questioning positions the user as a reflective expert ("tell me your general principle"); PI questioning positions the user as a validator of the agent's hypothesis ("I think I see a pattern—is this right?"). These framings have implications for perceived control and trust.

? classified proactive AI behaviors into perception, decision support, and task execution, finding that proactive agents improve task performance but can reduce user attention and perceived control when the agent acts without verification. **?** identified that LLM-powered voice assistants frequently produce expectation mismatches that erode trust over time. PI questioning is particularly exposed to this risk: if the agent's inferred pattern is wrong, confirmation requests can feel presumptuous. Our user study examines whether this theoretical concern manifests in practice.

3. An Active Questioning Agent for Household Rearrangement

3.1. Task Formulation

We consider a household rearrangement scenario defined by a tuple $(R, \mathcal{C}, \mathcal{O}_s, \mathcal{O}_u)$, where R is the room type, $\mathcal{C} = \{c_1, \ldots, c_M\}$ is the set of available receptacles (shelves, drawers, cabinets, etc.), and $\mathcal{O}_s$, $\mathcal{O}_u$ are the sets of seen and unseen objects, respectively. The agent may ask at most B questions (the *question budget*) before placing all objects.

Seen objects are visible to both user and agent during the interaction; unseen objects must be placed based on preferences generalized from the dialogue. This seen/unseen split provides a principled measure of generalization: a strategy that extracts transferable category rules should outperform one that only resolves individual placements on unseen objects, even when the two strategies perform similarly on seen objects.

3.2. Structured Preference State

Rather than retaining only raw dialogue history, the agent maintains a structured state $\mathcal{S}_t$ that separates different types of learned preference information:

- **Confirmed actions** $\mathcal{A}^+ = \{(o_i, c_j)\}$: specific object-receptacle pairs confirmed by the user.
- **Confirmed preferences** $\mathcal{P}^+ = \{(h_k, \mathcal{O}_k, c_k)\}$: category-level rules, where h_k is a natural-language hypothesis (e.g., "electronic accessories"), $\mathcal{O}_k$ is the set of covered objects, and c_k is the associated receptacle.
- **Negative evidence** $\mathcal{A}^-$, $\mathcal{P}^-$: placements and preferences explicitly rejected.
- **Unresolved objects** $\mathcal{U}_t \subseteq \mathcal{O}_s$: seen objects without a confirmed placement.
- **Dialogue history** $\mathcal{Q}_t = \{(q_i, a_i)\}_{i=1}^{t}$: the full question-answer record.

This representation enables the agent to track which objects remain unresolved and which category rules have been established, directly informing which questioning strategy and which specific question to use next. The structured state also facilitates principled placement planning after the question budget is exhausted (Section 3.4).

3.3. Three Active Questioning Strategies

We define three strategies that differ in target granularity, prerequisite state, and the cognitive demand they place on users:

3.3.1. Action-Oriented (AO) Questioning

AO targets a single unresolved object $o \in \mathcal{U}_t$ and asks directly about its placement (e.g., "Where should the wireless speaker go?"). The user's answer yields a confirmed action $(o, c) \in \mathcal{A}^+$. AO questions are concrete and low-effort for users—the correct answer is an object and a location—but each question resolves at most one seen object with no direct generalization to unseen objects. The proposer selects the target object by priority: (1) an object close to the boundary of a known preference category, offering the possibility of rule extension; (2) a representative object from a common household category; (3) any remaining unresolved object.

3.3.2. Preference Eliciting (PE) Questioning

PE asks about an organizing principle rather than a specific object (e.g., "How do you usually organize electronic accessories?"). The proposer generates a *hypothesis*—a short category label—and identifies the seen objects most plausibly belonging to this category. The user's answer yields a confirmed preference $(h, \mathcal{O}, c) \in \mathcal{P}^+$ that can resolve multiple objects simultaneously and generalize to unseen objects. PE places higher cognitive demands on users, who must reflect on their organizing habits at an abstract level rather than pointing to a specific location.

3.3.3. Preference Induction (PI) Questioning

PI leverages the pattern of confirmed actions in $\mathcal{A}^+$ to formulate a hypothesized rule (e.g., "I've noticed you place books and magazines on the display shelf—is that generally true for reading materials?"). The user confirms, refines, or rejects the hypothesis. When confirmed, PI converts implicit behavioral patterns into explicit preference rules, enabling generalization. PI requires sufficient prior evidence and may feel presumptuous when the agent's inference is incorrect.

3.4. Placement Planning After the Question Budget

After the question budget is exhausted, a final placement planner assigns all remaining objects (seen and unseen) to receptacles. For seen objects, the planner matches each unresolved object against confirmed preferences and confirmed actions, with confirmed preferences taking priority only when the object clearly belongs to the preference's category. For unseen objects, the planner applies a three-step inference:

Table 1
Comparison of three active questioning strategies across key dimensions.

	Action-Oriented	Preference Eliciting	Preference Induction
Target	Single object	Object category	Observed pattern
Example question	"Where should the speaker go?"	"How do you organize electronics?"	"Books always go on the shelf—correct?"
Information grain	Fine (1 object)	Coarse (N objects)	Coarse (N objects)
Prerequisite state	Unresolved objects	None	$\geq$3 confirmed actions
User cognitive demand	Low	Medium	Medium–High
Generalization potential	None (seen only)	High (seen + unseen)	High (seen + unseen)
User control framing	User as authority	User as reflective expert	User as validator

1. **Confirmed preference match**: if the unseen object belongs to a confirmed preference category h_k with an associated receptacle c_k, assign directly.

2. **Semantic analogy**: if no preference matches, find the most semantically similar confirmed action and assign the same receptacle.

3. **Frequency fallback**: assign the receptacle that appeared most frequently in confirmed actions.

Step 1 is the primary mechanism by which PE and PI questioning benefit unseen objects, making the extraction of explicit receptacle information from category answers critical to overall performance.

3.5. Implementation

The agent is implemented as a modular LLM pipeline using Qwen3 (served locally via Ollama). Each strategy has a dedicated proposer that enforces strategy-specific constraints: PE proposers generate category hypotheses and identify covered objects; AO proposers apply the priority selection described above. A state update module interprets natural-language answers and populates the structured preference state. We use structured output schemas (Pydantic models) throughout to ensure consistent parsing.

4. Study 1: Characterizing Strategy Performance Through Simulation

Study 1 aims to establish controlled performance baselines for each strategy and to characterize how their accuracy profiles differ across question budgets. By using LLM-simulated users (oracles), we can run experiments at a scale that would be infeasible with human participants, enabling systematic comparison across 102 household episodes and multiple budget values.

4.1. Dataset

We constructed a dataset of 102 household episodes spanning three room types: living rooms (34 episodes), bedrooms (34 episodes), and kitchens (34 episodes). Each episode contains 12 seen objects, 12 unseen objects, and 5–8 receptacles appropriate to the room type. Critically, each episode includes *annotator notes* describing the simulated user's organizing principles (e.g., "Small entertainment electronics go to the side drawer; books and magazines always go on the display shelf"). These notes define the oracle's knowledge base without being directly accessible to the agent.

4.2. Simulated User Oracle

The oracle answers the agent's questions based on ground-truth placements and annotator notes. It is instructed to behave as a natural household user: answers are concise, use exact receptacle names, and stay focused on the question asked. To maintain multi-turn consistency, the oracle receives the most recent five question-answer pairs and is instructed not to contradict prior answers. For preference-type questions, the

oracle provides one primary receptacle as the main organizing rule, avoiding room-level descriptions (e.g., "the bedside table" rather than "in the bedroom").

4.3. Hypotheses

Based on the theoretical framing in Section 3.3, we formulate five a-priori hypotheses:

H1 Active questioning will significantly outperform no-question baselines on both seen and unseen accuracy, confirming that user-provided information is necessary for accurate placement.

H2 Structured questioning strategies (AO, PE, PI) will outperform unstructured LLM questioning, confirming that explicit state tracking and strategy-specific proposers add value beyond raw language model capability.

H3 PE will achieve higher unseen accuracy than AO at equivalent budgets, because category-level rules generalize directly to unseen objects while object-level confirmed actions support only indirect analogy.

H4 The accuracy gap between PE and AO will widen with increasing budget, as each additional PE question compounds the number of resolved objects and extended rules.

H5 AO will provide faster initial accuracy gains for seen objects at low budgets ($B \leq 2$), as its deterministic object-resolution does not depend on hypothesis quality.

4.4. Experimental Conditions

We evaluate five conditions across question budgets $B \in \{0, 2, 4, 6\}$:

1. **No Question** ($B = 0$): The LLM assigns placements using only object and receptacle names, without any user interaction. This establishes the lower bound.
2. **Raw LLM**: An unstructured LLM freely generates questions and interprets answers without structured state tracking or strategy-specific prompting. This tests whether the agent architecture adds value beyond raw LLM capability (H2).
3. **AO Only**: The agent uses only Action-Oriented questioning throughout.
4. **PE Only**: The agent uses only Preference Eliciting questioning throughout.
5. **PI Cold-Start**: The agent uses only Preference Induction questioning without pre-seeded evidence, testing PI's ability to bootstrap from no prior confirmed actions.

4.5. Measures

We report two accuracy metrics across all conditions and budgets:

- **Seen accuracy**: fraction of seen objects assigned to the ground-truth receptacle.
- **Unseen accuracy**: fraction of unseen objects assigned to the ground-truth receptacle.

The gap between seen and unseen accuracy for each strategy indicates the extent to which that strategy's acquired knowledge is specific to the queried objects (high gap) or transfers to novel objects (low gap).

4.6. Results

4.6.1. No-Question Lower Bound

Across all 102 episodes at $B = 0$, the LLM achieves seen accuracy 0.427 and unseen accuracy 0.426 (Table 2). The near-identical scores confirm that without user interaction, the LLM has no informational advantage for objects it has "seen"—both are resolved purely from object name semantics. This establishes the baseline against which all questioning strategies are compared.

4.6.2. Placement Accuracy Across Strategies

Table 2 presents accuracy across all conditions and budgets.

Table 2
Study 1 placement accuracy by strategy and question budget (10-episode sample, seed=42). Seen = fraction of seen objects correctly placed; Unseen = fraction of unseen objects correctly placed.

	Budget = 2		Budget = 4		Budget = 6	
Strategy	Seen	Unseen	Seen	Unseen	Seen	Unseen
No Question	.427	.426	—	—	—	—
Raw LLM	.392	.442	.525	.517	.592	.525
AO Only	.533	.392	.583	.433	.700	.525
PE Only	.552	.531	.708	.646	.823	.719
PI Cold-Start	[TODO: run]	[TODO: run]	[TODO: run]	[TODO: run]	[TODO: run]	[TODO: run]

4.6.3. Hypothesis Evaluation

H1 supported: User-provided information proves necessary for accurate placement. Without any user interaction ($B = 0$), the agent achieves only 42.7% accuracy—a ceiling imposed by what can be inferred from object and receptacle names alone. Every active condition improves substantially on this baseline: at $B = 6$, PE reaches 82.3% seen accuracy, and even unstructured Raw LLM questioning reaches 59.2%. The need for user input cannot be substituted by LLM common sense (H1 confirmed).

H2 supported: Strategy-specific question design and structured state tracking provide measurable benefits over free-form dialogue. At $B = 6$, both AO (70.0%) and PE (82.3%) substantially outperform Raw LLM (59.2%) on seen accuracy. The advantage is more pronounced for unseen objects: PE achieves 71.9% versus Raw LLM's 52.5%. The gap confirms that maintaining explicit preference records—rather than simply accumulating dialogue history—allows the agent to act more reliably on what users say (H2 confirmed).

H3 supported: Category-level rules generalize to novel objects in ways that object-level placements cannot. At $B = 6$, PE unseen accuracy (71.9%) substantially exceeds AO (52.5%)—a gap of 19.4 percentage points. When the agent learns that "electronic accessories go on the media shelf", it can place any novel object in that category directly; when it only knows that "the remote control goes on the media shelf", it must rely on approximate semantic analogy for everything else. Seen accuracy shows a smaller but consistent PE advantage (82.3% vs. 70.0%), reflecting that AO's direct object-resolution is effective for queried objects but leaves the rest underspecified (H3 confirmed).

H4 supported: The information value of category-level questions compounds with each additional interaction. At $B = 2$, PE and AO differ by only 1.9 percentage points on seen accuracy (55.2% vs. 53.3%). By $B = 6$, that gap has grown to 12.3 points seen and 19.4 points unseen. Each PE question covers a category of 2–3 objects on average; the rules established early in the dialogue continue to resolve new objects throughout—an effect that accumulates with budget in a way that AO's one-object-per-question approach cannot match (H4 confirmed).

H5 partially supported: Direct object queries deliver reliable early returns but fail to build transferable knowledge. At $B = 2$, AO seen accuracy (53.3%) is close to PE (55.2%), consistent with H5's prediction of comparable early gains for seen objects. However, AO's unseen accuracy (39.2%) already trails PE (53.1%) substantially at this budget, indicating that AO's reliability at low budget comes entirely from resolved objects—not from any knowledge that carries over to the unseen set. H5 is partially confirmed: the parity holds for seen accuracy but not overall (H5 partially confirmed).

4.6.4. Implications for Study 2

The simulation results establish that AO and PE represent meaningfully distinct strategies: PE excels at generalization, AO at predictability. Whether these performance differences translate into differences in user experience—and whether real users answer PE's abstract questions as reliably as the oracle does—are questions that require human participants to answer. Study 2 is designed precisely to test these

extrapolations. PI will be included as a Study 2 condition pending the completion of cold-start simulation results; its addition would allow a three-way experiential comparison.

4.7. Limitations of Simulated Evaluation

The oracle provides consistent, well-formed answers using exact receptacle names—a best-case scenario. Real users may give ambiguous, incomplete, or multi-part answers, especially in response to PE's abstract questions. The simulation does not model frustration, hedging, or changing mental representations during the interaction. Study 2 directly addresses these limitations.

5. Study 2: How Questioning Strategies Affect User Experience

[TODO: This section presents the user study design and results. The subsections below provide the planned structure; content will be completed after data collection.]

5.1. Research Questions and Hypotheses

Study 2 investigates the experiential dimensions of questioning strategy that are not accessible through simulation. Guided by the theoretical framing in Section 2 and the simulation findings of Study 1, we examine:

RQ3. How do strategies differ in perceived control, cognitive load, trust, and satisfaction?

RQ4. Does organizing expertise moderate strategy effectiveness?

We formulate additional hypotheses:

H6 PE questioning will produce lower perceived control than AO, because abstract category questions reduce users' sense of moment-to-moment agency over placement decisions.

H7 AO questioning will produce higher cognitive load ratings at higher budgets than PE, because resolving objects one-by-one becomes tedious relative to category-level resolution.

H8 Users with high organizing expertise will prefer PE (able to articulate abstract rules), while users with low expertise will prefer AO (concrete, low cognitive demand).

5.2. Study Design

[TODO: Within-subject design. Each participant experiences 2–3 strategy conditions (counterbalanced using a Latin square). Independent variable: questioning strategy (AO / PE / Combined). Question budget fixed at $B = 6$ to match Study 1's highest-budget condition.]

5.3. Participants

[TODO: Target: 24–30 participants. Recruitment via university participant pool and community posting. Inclusion criteria: age 18–65, resident of a private home for at least 6 months. Organizing expertise assessed via 5-item self-report scale ("I have specific rules for where household items belong", 1–5 Likert). IRB approval obtained from [institution].]

5.4. Task and Procedure

[TODO: Participants interact with the agent through a web-based chat interface to organize a virtual room (12 seen + 12 unseen objects, same room types as Study 1). Each session: self-introduction + organizing expertise questionnaire → condition 1 interaction (budget=6) → post-condition questionnaire → repeat for conditions 2 and 3 → comparative preference interview. Three distinct room types used across conditions to minimize learning effects. Condition order counterbalanced.]

5.5. Interface

[TODO: Web chat interface. Agent presents questions; user types free-text responses. Room state displayed as a labelled diagram showing current placements in real time. All interactions, timestamps, and intermediate agent state logged. Pilot test with 3 users to verify interface usability.]

5.6. Measures

5.6.1. Task Performance

Seen accuracy and unseen accuracy (same definition as Study 1), enabling direct comparison with simulation results. Objects per question (information efficiency).

5.6.2. User Experience Questionnaire

[TODO: Post-condition questionnaire: NASA-TLX (cognitive load, 6 subscales); perceived control (5 items adapted from ?); trust in agent (5 items adapted from ?); satisfaction (1 item, 7-point scale). Comparative preference interview: "Which style did you prefer and why?", "Did you feel in control?", "Were any questions confusing?"]

5.6.3. Interaction Logs

[TODO: Response length per answer, answer hesitation (time from question display to first keystroke), number of follow-up clarifications requested.]

5.7. Analysis Plan

[TODO: Quantitative: linear mixed-effects models with strategy as within-subject factor and organizing expertise as continuous covariate, allowing strategy $\times$ expertise interaction to test H8. Random intercept per participant. Qualitative: thematic analysis of interview transcripts using reflexive coding (?); two coders, Cohen's κ reported.]

5.8. Results

5.8.1. Task Performance with Real Users

[TODO: Compare seen and unseen accuracy per strategy. Expected: PE $>$ AO for unseen accuracy, consistent with H3. Real users may give less precise answers, narrowing the gap. Raw LLM condition repeated if feasible.]

5.8.2. User Experience Across Strategies

[TODO: Report mixed-effects results for each outcome measure. Expected: PE lower perceived control (H6), AO higher cognitive load at $B = 6$ (H7). Present effect sizes; discuss practical significance alongside statistical significance.]

5.8.3. The Role of Organizing Expertise

[TODO: Report strategy $\times$ expertise interaction effects. Expected: experts prefer PE; novices prefer AO (H8). Illustrate with representative quotes from interviews.]

5.8.4. Interaction Patterns and Breakdowns

[TODO: Qualitative themes: when do users give multi-receptacle answers to PE questions? When do users reject PI hypotheses and why? What conversational repair strategies emerge? How do users express uncertainty? These patterns extend the simulation findings by exposing real user behaviors that the oracle cannot replicate.]

6. Discussion

6.1. Simulation Versus Real User Behavior: Where the Gap Lies

[TODO: Study 1 establishes that PE's category-level rules dramatically improve generalization in an ideal oracle environment. Study 2 will reveal whether this advantage persists when users give ambiguous answers, answer at varying levels of specificity, or misinterpret

abstract questions. We anticipate that the PE–AO accuracy gap will narrow with real users—the interesting question is *by how much* and *which user characteristics* determine the size of the gap. If expert organizers preserve PE's accuracy advantage while novices do not, this directly supports an adaptive strategy recommendation.]

6.2. The Accuracy–Agency Trade-off

[TODO: Discuss the tension between information efficiency (PE) and user agency (AO). Category-level questions are more efficient but implicitly ask the user to accept the agent's framing of their organizing categories—which may not match their actual mental model. Concrete object questions are less efficient but position the user as the definitive authority. This trade-off mirrors findings in conversational recommendation (?) where personalizing the interaction modality to user preferences improved both outcomes and satisfaction.]

The trade-off also has a temporal dimension: at low budgets ($B \leq 2$), AO and PE are nearly equivalent on seen accuracy. The accuracy cost of choosing AO over PE is small when only a few questions are permitted. Designers who prioritize user agency can reasonably default to AO for short interactions—using PE only when a larger question budget is available and the generalization benefit justifies asking users to reason more abstractly about their preferences.

6.3. Theoretical Contributions

[TODO: This work contributes three main theoretical claims: (1) The granularity of preference queries—not just their presence—determines the type and generalizability of learned preferences; the AO/PE/PI taxonomy captures this granularity across a dimension not previously formalized in the literature. (2) Structured state representation (confirmed actions and preferences as typed records) provides consistent accuracy benefits over raw dialogue history retention, suggesting that intermediate representation—not just language model scaling—is important for preference-learning agents. (3) Organizing expertise is a theoretically principled moderator of questioning strategy effectiveness: it predicts users' ability to respond usefully to abstract category questions. This extends the finding of ? that user experience level moderates LLM interaction quality into the domain of preference elicitation.]

6.4. Design Recommendations

Based on the simulation evidence from Study 1 and the predicted findings of Study 2, we offer four recommendations for designers of conversational agents that learn user preferences:

1. **Open with category-level questions when budget allows.** PE's compounding accuracy advantage is maximized over more questions. An agent with a budget of 4 or more should begin with PE to establish broad rules, then fall back to AO for remaining ambiguities.

2. **Match questioning granularity to user expertise.** Users who struggle to articulate organizing principles in category-level terms should be offered the more concrete AO style; users who readily express general rules stand to benefit more from PE's higher information yield. A brief initial exchange—or a simple self-report item—can inform this routing.

3. **Maintain structured preference records rather than raw dialogue.** Confirmed actions and preferences with explicit receptacle associations substantially outperform free-form dialogue retention for downstream placement. The gains are especially large for unseen objects, where the agent cannot rely on direct recall of a prior answer.

4. **Give users explicit control over questioning style.** Because strategies differ in how much agency users feel over placement decisions, offering a brief framing at the start of the interaction ("I can ask you about specific items, or about your general organizing habits—which would you prefer?") respects user autonomy and may improve satisfaction without sacrificing accuracy.

6.5. Limitations

Four limitations bound the scope of the current findings. First, the simulated oracle represents a best-case user: consistent, precise, and fluent with exact receptacle names. Real users give more varied responses, and Study 2 will begin to characterize that variability—but even a lab user study captures a narrow demographic and cultural context. The norms governing "where things belong" vary substantially across households and cultures; whether our strategy taxonomy generalizes across those norms is an open question.

Second, the evaluation covers three room types (living room, bedroom, kitchen) representing prototypical residential spaces. Strategies may behave differently in more specialized environments (home offices, workshops, shared spaces) or in rooms with ambiguous or overlapping receptacle categories.

Third, all reported results use a single LLM backend (Qwen3). The proposer and state-update modules depend on the model's ability to generate well-formed structured outputs; performance may vary with model capability, temperature settings, or prompt design. Replication with alternative models is needed before generalizing the quantitative findings.

Fourth, both the simulation and the planned user study are single-session. Household preferences are not static: users learn, reorganize, and develop new habits over time. A longitudinal evaluation—where the agent refines its preference model across multiple sessions—would provide a more ecologically valid test of each strategy's long-term utility.

6.6. Future Directions

Several extensions follow naturally from this work. Adaptive policy selection—dynamically choosing among AO, PE, and PI based on accumulated uncertainty estimates (e.g., entropy reduction per expected question)—is a principled next step that our structured state representation already supports. Deployment on a physical robot would introduce perception noise and real-world object variability, testing the limits of preference generalization in a way that simulation cannot. Longitudinal multi-session studies would examine how learned preferences persist and evolve. Finally, pre-registered, larger-scale user studies would provide higher-powered tests of the expertise moderation hypothesis (H8) and the design recommendations derived from it.

7. Conclusion

We investigated how an intelligent agent should ask for user preferences when assisting with household rearrangement. Grounding the investigation in three qualitatively distinct questioning strategies—Action-Oriented, Preference Eliciting, and Preference Induction—we showed that the *how* of asking has consequences that extend far beyond information efficiency.

Simulation across 102 household episodes demonstrated that structured questioning strategies substantially outperform both no-question baselines (which achieve only 42.6% accuracy) and unstructured LLM querying. Category-level Preference Eliciting questioning nearly doubles unseen-object accuracy at a budget of 6 questions (71.9% vs. 42.6% baseline), because category rules transfer directly to novel objects while object-level confirmed actions support only indirect analogy. The accuracy advantage of PE over Action-Oriented questioning grows with question budget, confirming the compounding value of category-level rules.

[TODO: Study 2 will further reveal how these performance differences translate to user experience: which strategy users prefer, when they feel in control, and how organizing expertise shapes those judgments. [Complete after data collection.]]

Together, these findings establish that active questioning is necessary for accurate household rearrangement assistance, that structured state representation adds consistent value over unstructured dialogue, and that strategy selection should be informed by user characteristics rather than treated as a fixed design choice. The question of how to ask—not just what to ask—fundamentally shapes both the quality of learned preferences and the user's experience of the interaction. Addressing that question systematically is a step toward conversational agents that learn from users not just more efficiently, but in ways that respect their organizing knowledge and reinforce their sense of control over their own homes.

A. Prompt Templates

[TODO: Include key prompt templates for each strategy's proposer, state update interpreter, and final placement planner. Representative examples of oracle answers for each strategy type. This supports reproducibility.]

B. Dataset Statistics

[TODO: Table: distribution of objects per receptacle across room types. Table: receptacle set size by room. Examples of annotator notes per room type. Inter-annotator agreement if applicable.]

C. User Study Materials

[TODO: Pre-task questionnaire (demographics, organizing expertise scale). Post-condition questionnaire (NASA-TLX, perceived control, trust, satisfaction). Semi-structured interview protocol. Informed consent form. Counterbalancing schedule.]